



Series · Mark Scheme

Core Pure 2 Chapter 2: Series. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Maclaurin series from first principles

- 1 $f(0) = 1, f'(0) = 3, f''(0) = 9$ B1; $e^{3x} \approx 1 + 3x + \frac{9}{2}x^2$ B1.
- 2 $f' = -2\sin 2x, f'' = -4\cos 2x, f''' = 8\sin 2x, f^{(4)} = 16\cos 2x$ M1; $f(0) = 1, f''(0) = -4, f^{(4)}(0) = 16$, odd derivatives zero A1; $1 - 2x^2 + \frac{2}{3}x^4$ A1.
- 3 $f' = \frac{2}{1+2x}, f'' = \frac{-4}{(1+2x)^2}, f''' = \frac{16}{(1+2x)^3}$ M1; $f(0) = 0, f'(0) = 2, f''(0) = -4, f'''(0) = 16$ A1; $2x - 2x^2 + \frac{8}{3}x^3$ A1.
- 4 $f' = \sec^2 x, f'' = 2\sec^2 x \tan x$ M1; $f''' = 4\sec^2 x \tan^2 x + 2\sec^4 x$ M1; $f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = 2$ A1; $x + \frac{2}{3}x^3 = x + \frac{1}{3}x^3$ A1.
- 5(a) $f'(x) = \cos x e^{\sin x}$ B1; product rule: $f''(x) = -\sin x e^{\sin x} + \cos^2 x e^{\sin x} = (\cos^2 x - \sin x) e^{\sin x}$ B1.
- 5(b) Product rule M1; $f'''(x) = (-2\sin x \cos x - \cos x) e^{\sin x} + (\cos^2 x - \sin x) \cos x e^{\sin x}$ A1.
- 5(c) $f(0) = 1, f'(0) = 1, f''(0) = 1$ B1; $f'''(0) = -1 + 1 = 0$ B1; $e^{\sin x} \approx 1 + x + \frac{1}{2}x^2$ (the x^3 term is zero) B1.
- 5(d) $\int_0^{0.2} (1 + x + \frac{1}{2}x^2) dx = [x + \frac{1}{2}x^2 + \frac{1}{6}x^3]_0^{0.2}$ M1; $= 0.221$ A1.

Section B – Standard series and binomial

- 6 Replace x by $-2x$ in the series for e^x M1; $1 - 2x + 2x^2 - \frac{4}{3}x^3$ A1.
- 7 $(-3x) - \frac{(-3x)^2}{2} + \frac{(-3x)^3}{3}$ M1; $-3x - \frac{9}{2}x^2 - 9x^3$ A1; valid for $-\frac{1}{3} \leq x < \frac{1}{3}$ B1.
- 8 $1 + \frac{1}{2}(4x) + \frac{\frac{1}{2}(-\frac{1}{2})}{2!}(4x)^2 + \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!}(4x)^3$ M1; $1 + 2x - 2x^2 + 4x^3$ A1; valid for $|x| < \frac{1}{4}$ B1.
- 9 $(4-x)^{-\frac{1}{2}} = \frac{1}{2}(1 - \frac{x}{4})^{-\frac{1}{2}}$ M1; $(1 - \frac{x}{4})^{-\frac{1}{2}} = 1 + \frac{1}{8}x + \frac{3}{128}x^2 + \frac{5}{1024}x^3$ M1; $\frac{1}{2} + \frac{1}{16}x + \frac{3}{256}x^2 + \frac{5}{2048}x^3$ A1; valid for $|x| < 4$ B1.
- 10(a) $\sqrt{4+x} = 2(1 + \frac{x}{4})^{\frac{1}{2}}$ M1; binomial expansion with $\frac{x}{4}$ M1; $2 + \frac{1}{4}x - \frac{1}{64}x^2 + \frac{5}{512}x^3$ A1; valid for $|x| < 4$ B1.
- 10(b) $\sqrt{4.04} \approx 2 + 0.01 - 0.000025 + 0.000000125 = 2.009975125$ M1; $\sqrt{4.04} = \frac{\sqrt{404}}{10} = \frac{2\sqrt{101}}{10}$, so $\sqrt{101} = 5\sqrt{4.04}$ M1; $\sqrt{101} \approx 10.049876$ A1.
- 10(c) $x = 97$ is outside the range of validity $|x| < 4$ (the series diverges) B1.

Section C – Combining series

- 11 Replace x by x^2 M1; $1 + x^2 + \frac{1}{2}x^4$ A1.
- 12 $(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3)(x - \frac{1}{6}x^3)$ M1; multiplying out and collecting terms up to x^3 M1; $x + x^2 + \frac{1}{3}x^3$ A1.
- 13 $\ln(1+x) - \ln(1-x)$ with $\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \dots$ M1; even powers cancel, giving $2x + \frac{2}{3}x^3 + \frac{2}{5}x^5$ A1; $\ln(1+x)$ needs $-1 < x \leq 1$ and $\ln(1-x)$ needs $-1 \leq x < 1$; both must hold, so $-1 < x < 1$ B1.
- 14 $\ln(1+x+x^2) = \ln(1-x^3) - \ln(1-x)$ M1; $\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4$ B1; $\ln(1-x^3) = -x^3 + \dots$ B1; $x + \frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4$ A1.
- 15(a) $\cos x = 1 + u$ with $u = -\frac{1}{2}x^2 + \frac{1}{24}x^4$ M1; $\ln(1+u) = u - \frac{1}{2}u^2 + \dots$ M1; $u^2 = \frac{1}{4}x^4 + \dots$ A1; $-\frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{8}x^4$ M1; $= -\frac{1}{2}x^2 - \frac{1}{12}x^4$ A1.
- 15(b) $\ln(\sec x) = -\ln(\cos x) \approx \frac{1}{2}x^2 + \frac{1}{12}x^4$ B1.



15(c) $\int_0^{0.5} \left(-\frac{1}{2}x^2 - \frac{1}{12}x^4\right) dx = \left[-\frac{1}{6}x^3 - \frac{1}{60}x^5\right]_0^{0.5} M1; = -0.020833 - 0.000521 M1; = -0.0214 A1.$

Roar-level challenge

16 $e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3$ and $(1+x)^{-1} = 1 - x + x^2 - x^3 B1$; multiplying and collecting terms $M1$; $f(x) = 1 + x + x^2 + \frac{1}{3}x^3$
 $A1$; valid for $-1 < x < 1$ (the e^{2x} series is valid for all x) $B1$; $\frac{x^2 + \frac{1}{3}x^3 + \dots}{x^2} \rightarrow 1 B1.$